HK IMO Training 2019-2020
Problem Set 29 Suggested Solutions

Problem 113. A positive integer is called *squarefree* if it has no square divisors larger than 1. For any positive integer n, define

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 p_2 \cdots p_k \text{ for some pairwise distinct primes } p_1, p_2, \ldots, p_k, \\ 0 & \text{if } n \text{ is not squarefree.} \end{cases}$$

Let $x_1, x_2, \ldots$ be a sequence of positive integers and let $y_n = \sum_{j|n} x_j$ for each positive integer n. Prove that

$$x_n = \sum_{j|n} \mu(j) y_{\frac{n}{j}}$$

for each positive integer n.

Solution. First we show that

$$\sum_{j|m} \mu(j) = \begin{cases} 1 & \text{if } m = 1, \\ 0 & \text{if } m > 1, \end{cases}$$

For $m = 1$, the result is obvious. For $m > 1$, let $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ be the prime factorization of m. Now observe that in the sum $\sum_{j|m} \mu(j)$ the only nonzero terms come from $j = 1$ and from those divisors of m which are products of distinct primes. Hence

$$\sum_{j|m} \mu(j) = \mu(1) + \mu(p_1) + \mu(p_2) + \ldots + \mu(p_k) + \mu(p_1 p_2) + \ldots + \mu(p_{k-1} p_k) + \ldots + \mu(p_1 p_2 \cdots p_k)$$

$$= 1 + \binom{k}{1}(-1) + \binom{k}{2}(-1)^2 + \ldots + \binom{k}{k}(-1)^k = (1-1)^k = 0.$$

Now, for convenience, we use $x(n), y(m)$ for x_n, y_m respectively. Then

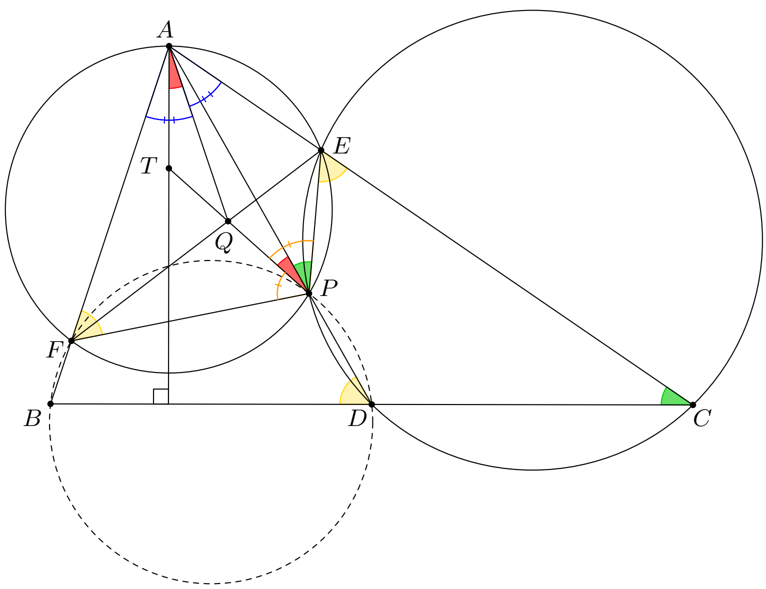
$$\sum_{j|n} \mu(j) y\left(\frac{n}{j}\right) = \sum_{j|n} \left(\mu(j) \sum_{k|\frac{n}{j}} x(k) \right) = \sum_{k|n} \left(x(k) \sum_{j|\frac{n}{k}} \mu(j) \right) = x(n)$$

because last inner sum is 0 unless $n = k$ by the result above. □

Remark. The problem above is the well-known *Möbius inversion formula*.

Problem 114. In a non-isosceles acute triangle ABC, D is the midpoint of BC. The points E and F lie on AC and AB, respectively, and the circumcircles of $\triangle CDE$ and $\triangle AEF$ intersect at P on AD. The interior angle bisector of $\angle FPE$ in $\triangle EFP$ intersects the line EF at Q. Prove that the tangent line to the circumcircle of $\triangle AQP$ at A is perpendicular to BC.

Solution.



Since C, D, P, E are concyclic, as well as A, E, P, F are concyclic, we have $\angle BDP = \angle PEC = \angle AFP$ and so B, D, P, F are concyclic. Since C, D, P, E are concyclic, we have $\dfrac{PE}{DC} = \dfrac{AE}{AD}$. Similarly, since B, D, P, F are concyclic, we have $\dfrac{PF}{BD} = \dfrac{AF}{AD}$. Thus, as $BD = DC$, we obtain $\dfrac{PF}{PE} = \dfrac{AF}{AE}$. On the other hand, since PQ is an angle bisector, we have $\dfrac{PF}{PE} = \dfrac{QF}{QE}$ and so $\dfrac{QF}{QE} = \dfrac{AF}{AE}$. which implies that AQ is the angle bisector of $\angle FAE$ in $\triangle FEA$.

Let T be the intersection of the line PQ and the line perpendicular to BC passing through A. Since AT is an altitude and AQ is angle bisector in $\triangle ABC$, we have $\angle TAQ = \dfrac{1}{2}(\angle ABC - \angle BCA)$. By angle chasing we have $\angle APE = \angle BCA$ and $\angle APF = \angle ABC$. As PQ is an angle bisector in FPE, we get $\angle QPA = \frac{1}{2}(\angle ABC - \angle BCA)$ and hence, we obtain $\angle TAQ = \angle QPA$ which implies that AT is tangent to the circumcircle of $\triangle AQP$ and we are done. $\square$

Problem 115. Let $P(x)$ be a non-constant polynomial with real coefficients such that all of its roots are real numbers. Suppose that there exists a polynomial $Q(x)$ with real coefficients such that

$$(P(x))^2 = P(Q(x)) \quad \text{for all real numbers } x.$$

Prove that all the roots of $P(x)$ are equal.

Solution. Let $P(x) = A(x - r_1)^{d_1} \cdot \ldots \cdot (x - r_k)^{d_k}$ where $r_1 < \cdots < r_k$. By considering the degrees of P and Q, we see that $2 \deg P = (\deg P)(\deg Q)$ and so $\deg Q = 2$ (since P is non-constant). Hence $Q(x) = ax^2 + bx + c$ for some real numbers a, b and c. Then the given equality can be written as

$$A^2 \prod_{j=1}^{k} (x_j - r_j)^{2d_j} = A \prod_{j=1}^{k} (ax^2 + bx + c - r_j)^{d_j}.$$

Therefore, for each j the roots of $ax^2 + bx + c - r$ are r_s and r_t for some $s, t \in \{1, \ldots, k\}$. On the other hand, the sum of the roots are $-b/a$ for every j. Thus, all the roots of $(P(x))^2$ can be paired in such a way that sum of the elements in each pair is the same. Let an r_1 be paired with r_s and an r_k be paired with r_t. Since $r_1 \leqslant r_t$, $r_s \leqslant r_k$ and $r_1 + r_s = r_k + r_t$, we see that $r_s = r_k$ and $r_t = r_1$. In other words, every r_1 has to be paired with an r_k and every r_k has to be paired with an r_1. Therefore, we obtain that $d_1 = d_k$. By induction it is easy to prove that $d_j = d_{k+1-j}$ for every j and all pairs are of the form $\{r_j, r_{k+1-j}\}$.

Consequently, for every m we have that $(c - r_m)/a$ is equal to $r_j r_{k+1-j}$ for some j. However, the numbers of the form $r_j r_{k+1-j}$ can attain at most $\left\lfloor \frac{k+1}{2} \right\rfloor$ distinct values and hence we get $k \leqslant \left\lfloor \frac{k+1}{2} \right\rfloor$ which implies $k \leqslant 1$. Therefore, all the roots of $P(x)$ are the same. $\square$

Problem 116. For positive integer k, let

$$R_n = \begin{cases} \{-k, -(k-1), \ldots, -1, 1, \ldots, k-1, k\} & \text{for } n = 2k, \\ \{-k, -(k-1), \ldots, -1, 0, 1, \ldots, k-1, k\} & \text{for } n = 2k+1. \end{cases}$$

A device consists of several balls and red or white ropes connecting some pairs of balls. A *labelling* is a colouring of each ball by one of the elements of R_n. We say that a labelling is

- *good* if colours of any two connected balls are different;
- *sensitive* if the colours of any two balls connected by white rope are different and the sum of colours of any two balls connected by red rope is not equal to 0.

Let $n \geqslant 3$ be a fixed integer. Suppose that any device which has a good labelling by R_n has also a sensitive labelling by R_m. Find the smallest possible value of m as a function of n.

Solution. The answer is $m = 2n - 1$.

Let us show that if a device has a good labelling by R_n then it has a sensitive labelling by R_m, where $m = 2n - 1$. In

$$R_m = R_{2n-1} = \{-(n-1), -(n-2), \ldots, -1, 0, 1, \ldots, n-2, n-1\}$$

there are n non-negative elements. Any good labelling of the device by these n non-negative elements will be also a sensitive labelling.

Now we construct a device which has a good labelling by R_n and has no sensitive labelling for any $m < 2n - 1$. First we define a grid $2n - 1$ rows and n columns, and place a ball into each cell. Then we connect any two balls belonging to the same row by a red rope and any two balls belonging to different rows and different columns by a white rope (there is no rope between any two balls from the same column). The device has a good labelling by R_n. Indeed, there are n distinct colours and if we colour all balls from the same column identically and balls from different columns differently then we get a good labelling. Suppose that the device has a sensitive colouring by R_m.

Case 1. There is a row all balls of which are differently coloured. Since all balls of this row are connected by red ropes, for each k at most one of the colours $\{-k, k\}$ is used. Therefore, the total number of elements in R_m with different absolute values should be at least n and consequently $m \geqslant 2n - 1$.

Case 2. Each row contains at least two identically coloured balls. Suppose that the repeated colour on some two rows are a and b. Since any two balls belonging to different rows and different columns are connected by a white rope we get $a \neq b$. Therefore, repeated colours of any two rows are different and the total number of different colours is at least $2n - 1$ and consequently $m \geqslant 2n - 1$. $\square$